

Series and Progressions

CSC 1105 Mathematics for Computer Science

Denish Azamuke, PhD

Office: CoCIS, Block B, Level 3

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Lecture Outline

- 1 Sequences, series, and summation notation
- 2 Arithmetic and geometric progressions
- 3 Special progressions and combined series
- 4 Taylor and Maclaurin series
- 5 Sequence approximation using interpolation

Learning Outcomes

By the end of this lecture, you should be able to:

- distinguish between a sequence, a progression, and a series;
- derive and apply formulae for arithmetic and geometric sums;
- simplify sums using special progressions;
- approximate functions using Taylor and Maclaurin polynomials;
- construct interpolation polynomials from discrete data points.

Part 2/2

Taylor S



Taylor Series



Taylor Series: Motivation

Computers often evaluate functions more easily when they are written as polynomials.

Idea

A differentiable function can be approximated near a point by a polynomial:

$$f(x) \approx c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots + c_k(x - a)^k.$$

Taylor and Maclaurin series provide a systematic way to find the coefficients.

Taylor Series

Suppose f is sufficiently differentiable near $x = a$. The Taylor series of f about a is

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \cdots$$

In summation notation,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

A truncated Taylor series is called a **Taylor polynomial**.

Maclaurin Series

A Maclaurin series is a Taylor series about $a = 0$.

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

General Form

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k.$$

Maclaurin Series for e^x

Example

Find the Taylor series approximation of e^x around $x = 0$ up to the fourth power. Estimate $e^{0.5}$ to five decimal places.

For $f(x) = e^x$,

$$f'(x) = f''(x) = f'''(x) = \cdots = e^x.$$

At $x = 0$,

$$f(0) = f'(0) = f''(0) = \cdots = 1.$$

Therefore,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

Which is the same as

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Approximating $e^{0.5}$

Using the fourth-degree Maclaurin polynomial,

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}.$$

Substitute $x = 0.5$:

$$\begin{aligned} e^{0.5} &\approx 1 + 0.5 + \frac{0.5^2}{2!} + \frac{0.5^3}{3!} + \frac{0.5^4}{4!} \\ &= 1.64844. \end{aligned}$$

Estimate

$e^{0.5} \approx 1.64844$

Exercise: Taylor and Maclaurin Series

① Find the Maclaurin expansion of:

a. $\sin x$

b. $\cos x$

c. $\frac{1}{1+x}$

d. $\sqrt{1+x}$

② Find a Taylor expansion and use it to estimate:

i. $\ln(1+x)$ for $\ln 2$ and $\ln 3$

ii. $\frac{1}{1-x}$ for $\frac{1}{0.7}$ and $\frac{1}{1.5}$

iii. e^{1+x^2} for e^3

Sequence Approximation

Sometimes we know only data points, not the rule generating them.

Example

$$(0, 0), (1, -1), (2, 3), (3, 17), (4, 47)$$

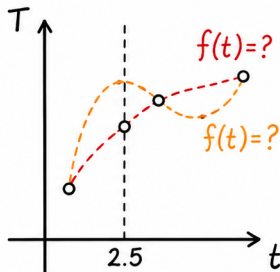
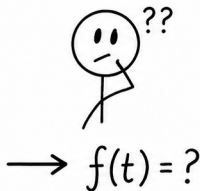
We may want to:

- estimate missing values;
- predict nearby values;
- find an approximate polynomial model;
- estimate rates of change by differentiating the polynomial.

Lagrange Polynomial Interpolation: Creates interpolation functions using Polynomials!



t	T
1	10
2	15
$\vdots$	$\vdots$
10	90



$$\text{Lagrange Polynomials} : \sum_{j=1}^n \left[y_j \prod_{\substack{k=1 \\ k \neq j}}^n \frac{x - x_k}{x_j - x_k} \right]$$

n = # of data points

j = term iteration

k = product notation
iteration counter

Lagrange Interpolation

Given data points

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n),$$

the Lagrange interpolating polynomial is

$$P_n(x) = \sum_{j=0}^n y_j L_j(x),$$

where the **Lagrange basis polynomial** is

$$L_j(x) = \prod_{\substack{i=0 \\ i \neq j}}^n \frac{x - x_i}{x_j - x_i}.$$

Key Idea

Each $L_j(x)$ is constructed so that $L_j(x_j) = 1$ and $L_j(x_i) = 0$ for $i \neq j$.

Lagrange Interpolation: Combined Form

Substituting $L_j(x)$ into

$$P_n(x) = \sum_{j=0}^n y_j L_j(x),$$

gives the equivalent form

$$P_n(x) = \sum_{j=0}^n y_j \prod_{\substack{i=0 \\ i \neq j}}^n \frac{x - x_i}{x_j - x_i}$$

Important

The polynomial $P_n(x)$ passes through every data point:

$$P_n(x_j) = y_j, \quad j = 0, 1, \dots, n.$$

Example: Lagrange Interpolation

Use Lagrange interpolation to estimate $f(2)$ from the data:

x	-2	-1	0	4
$f(x)$	-2	4	1	8

Thus, the four data points are

$$(-2, -2), \quad (-1, 4), \quad (0, 1), \quad (4, 8).$$

Recall that

$$P_3(x) = \sum_{j=0}^3 y_j L_j(x),$$

where

$$L_j(x) = \prod_{\substack{i=0 \\ i \neq j}}^3 \frac{x - x_i}{x_j - x_i}.$$

Example: Constructing the Lagrange Polynomial

For four data points, the polynomial is

$$P_3(x) = y_0L_0(x) + y_1L_1(x) + y_2L_2(x) + y_3L_3(x).$$

Explicitly,

$$P_3(x) = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}y_0 + \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)}y_1 + \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)}y_2 + \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)}y_3.$$

Example: Constructing the Lagrange Polynomial Continued

For the data points

$$(-2, -2), \quad (-1, 4), \quad (0, 1), \quad (4, 8),$$

substitute the values directly into the Lagrange formula:

$$\begin{aligned} P_3(x) = & (-2) \frac{(x - (-1))(x - 0)(x - 4)}{(-2 - (-1))(-2 - 0)(-2 - 4)} \\ & + 4 \frac{(x - (-2))(x - 0)(x - 4)}{(-1 - (-2))(-1 - 0)(-1 - 4)} \\ & + 1 \frac{(x - (-2))(x - (-1))(x - 4)}{(0 - (-2))(0 - (-1))(0 - 4)} \\ & + 8 \frac{(x - (-2))(x - (-1))(x - 0)}{(4 - (-2))(4 - (-1))(4 - 0)}. \end{aligned}$$

Example: Estimating $f(2)$

Substituting $x = 2$ gives

$$\begin{aligned} P_3(2) &= (-2) \frac{(3)(2)(-2)}{(-1)(-2)(-6)} \\ &\quad + 4 \frac{(4)(2)(-2)}{(1)(-1)(-5)} \\ &\quad + 1 \frac{(4)(3)(-2)}{(2)(1)(-4)} \\ &\quad + 8 \frac{(4)(3)(2)}{(6)(5)(4)}. \end{aligned}$$

Example: Estimating $f(2)$

Therefore,

$$\begin{aligned}P_3(2) &= (-2)(1) + (4)\left(-\frac{16}{5}\right) + (1)(3) + (8)\left(\frac{1}{5}\right) \\&= -2 - \frac{64}{5} + 3 + \frac{8}{5} \\&= -\frac{51}{5}.\end{aligned}$$

$$f(2) \approx P_3(2) = -10.2$$

Lagrange Example

Given the points

$$(0, 1), (2, 3), (4, 9), (6, 19),$$

construct the cubic Lagrange polynomial.

$$P_3(x) = L_0(x)y_0 + L_1(x)y_1 + L_2(x)y_2 + L_3(x)y_3.$$

After simplification,

$$P_3(x) = \frac{x^2}{2} + 1.$$

Using the Interpolation Polynomial

Using

$$P_3(x) = \frac{x^2}{2} + 1,$$

we estimate:

$$P_3(5) = \frac{25}{2} + 1 = 13.5,$$

$$P_3(7) = \frac{49}{2} + 1 = 25.5,$$

$$P_3(8) = \frac{64}{2} + 1 = 33.$$

Interpretation

The polynomial provides a compact rule for estimating values not directly listed in the data.

Revision Questions

- 1 Find the sum of the first n terms of $5, 8, 11, 14, \dots$
- 2 Find the sum to infinity of $12 + 6 + 3 + \frac{3}{2} + \dots$
- 3 Evaluate $\sum_{k=1}^n (3k^2 - 2k + 1)$.
- 4 Use a fourth-degree Maclaurin polynomial to approximate $\cos(0.2)$.
- 5 Construct the Lagrange polynomial for $(0, 2), (1, 3), (2, 6)$.

Questions?